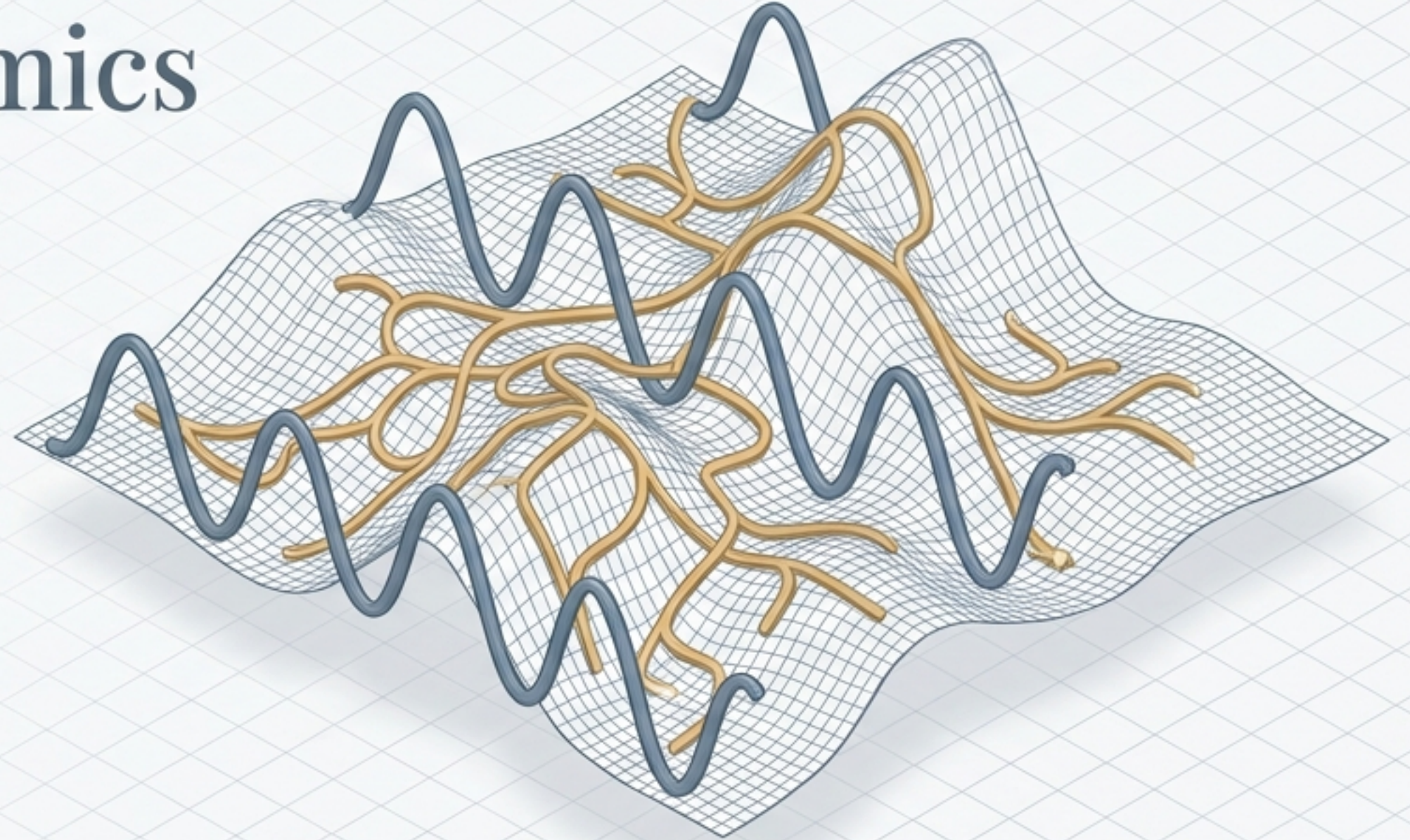


# From Wave Dynamics to Continuation Dynamics

Why Neural Oscillations May Be Realizations, Rather Than Explanations, of Cognitive Control

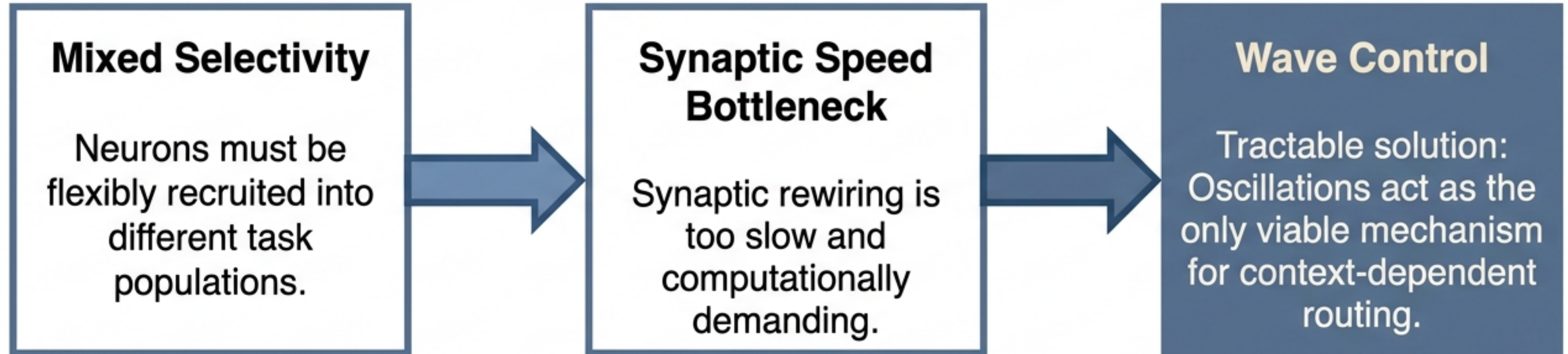
**Flyxion** | Independent Researcher





# The Prevailing Claim for Computational Necessity

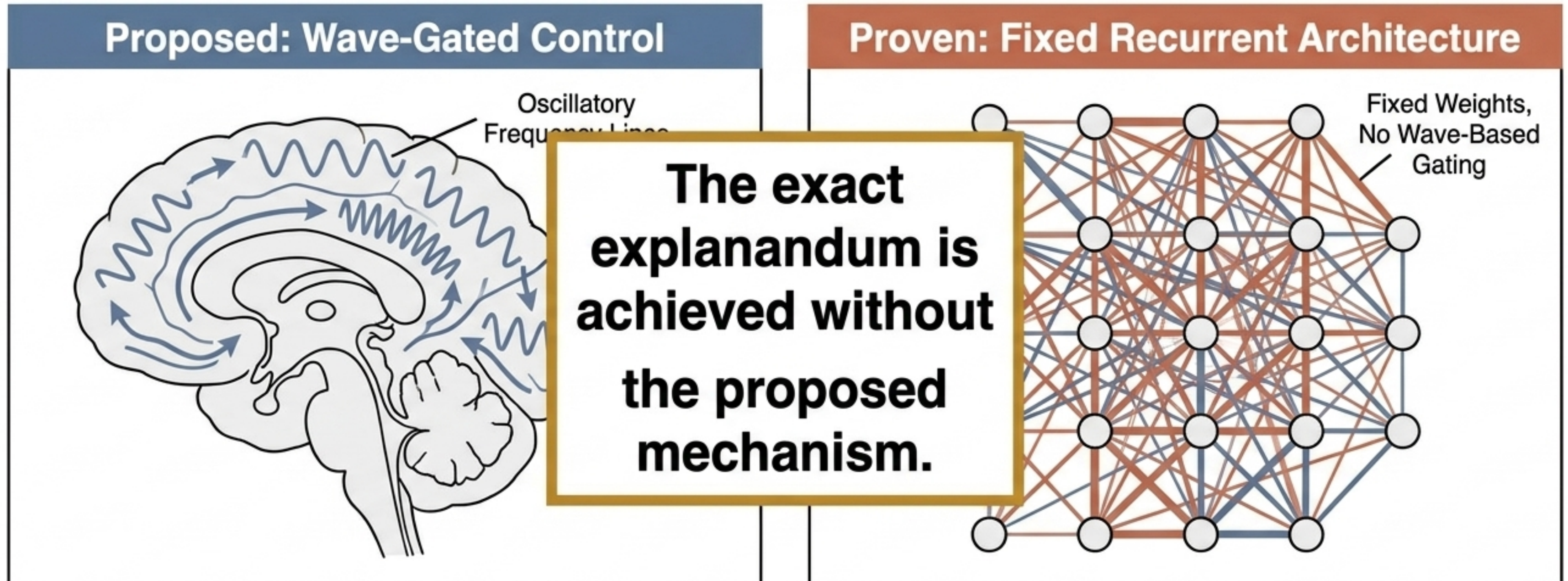
Miller et al. argue that spatially patterned oscillations (brain waves) are the unique solution to cortical mixed selectivity. Synaptic rewiring is deemed too slow, imposing an implausible computational demand.





# Fixed Architectures Already Solve the Control Problem

Mante et al. (2013) demonstrated that fixed recurrent neural networks (RNNs) produce context-dependent, sub-second routing without any wave-based gating mechanisms.

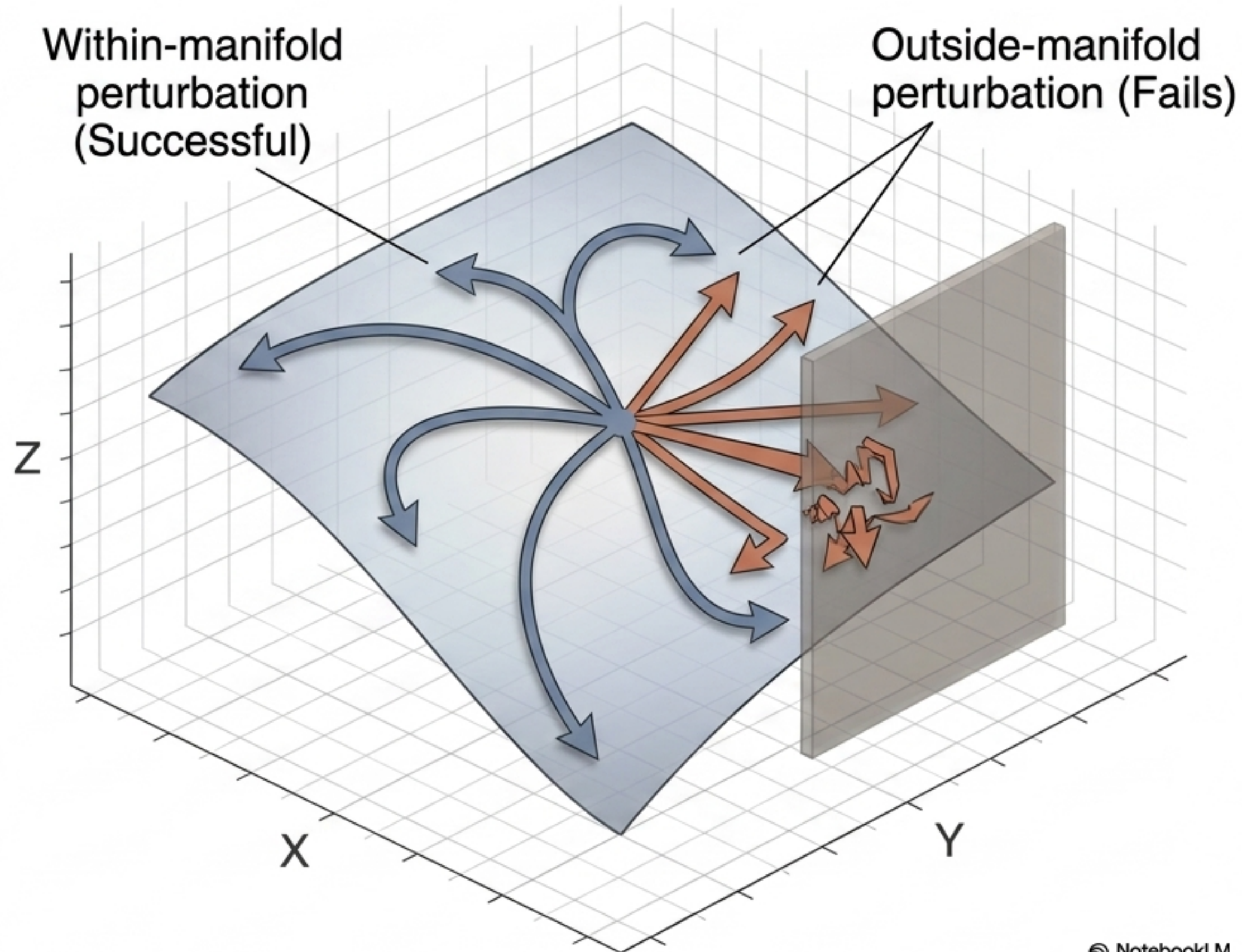




# Manifold Theory Already Predicts Short-Term Adaptation

For short-term perturbations, neural stability is strictly governed by the Intrinsic Manifold—the covariance structure reflecting fixed synaptic connectivity.

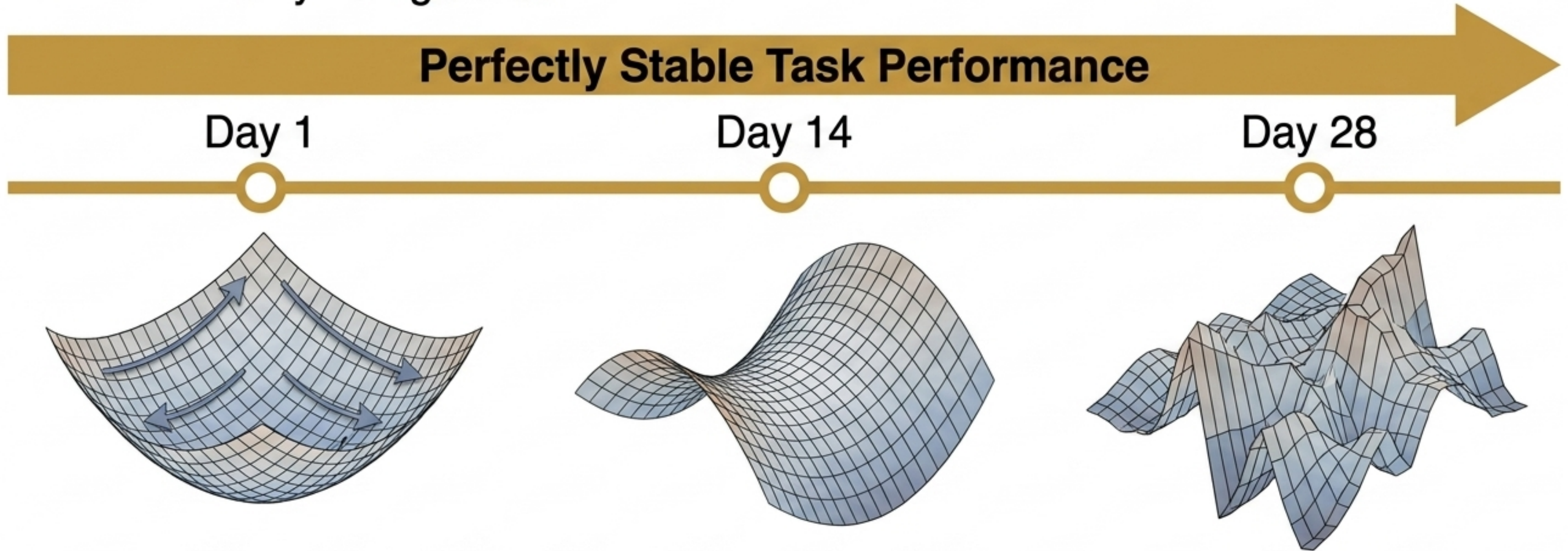
The repertoire is fixed; adaptation operates by reassignment, not expansion.





# The Explanatory Gap of Representational Drift

Driscoll et al. (2017) revealed that over weeks, individual neuronal relationships and the intrinsic manifold drastically reconfigure. If the geometric manifold shifts, what is actually being conserved?

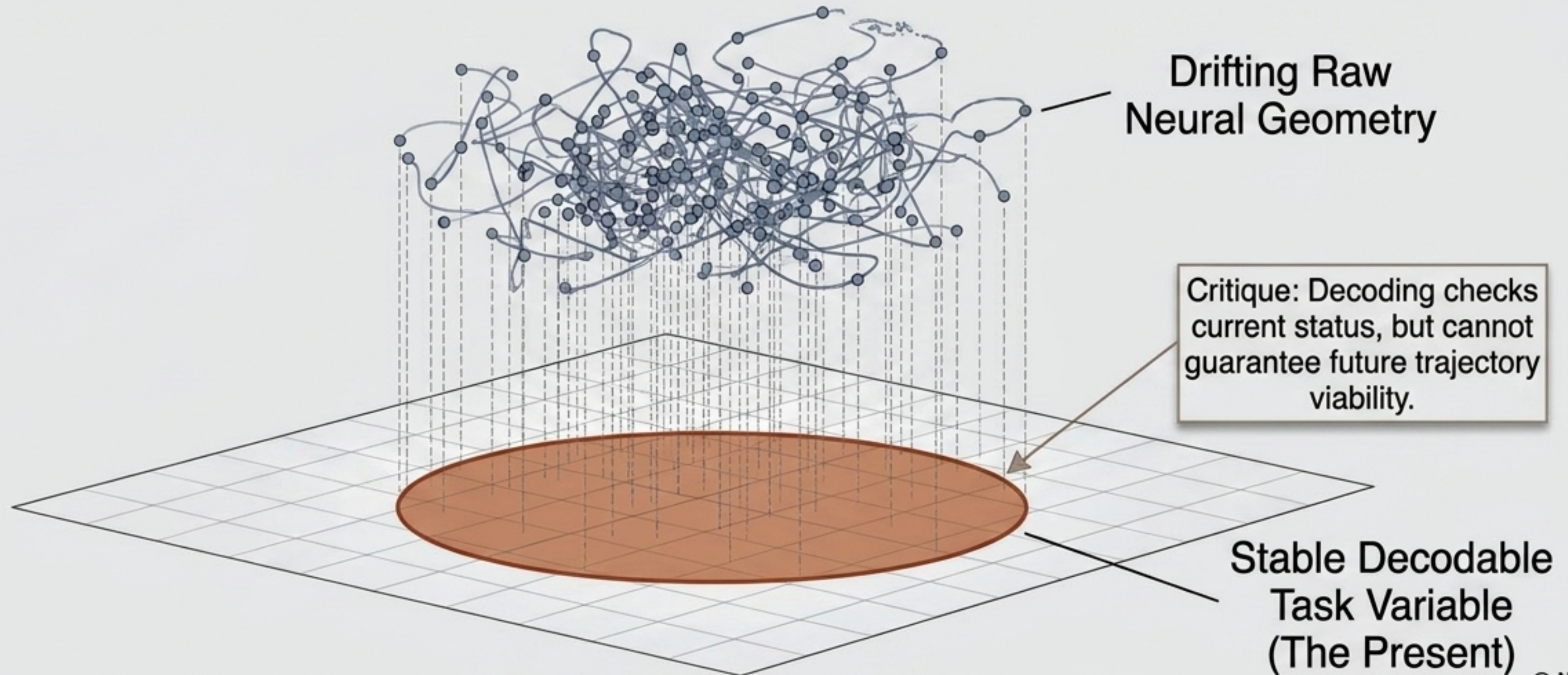


**Drifting Neural Geometry & Reconfigured Intrinsic Manifold**



# Stable Decodability is Only Half an Answer

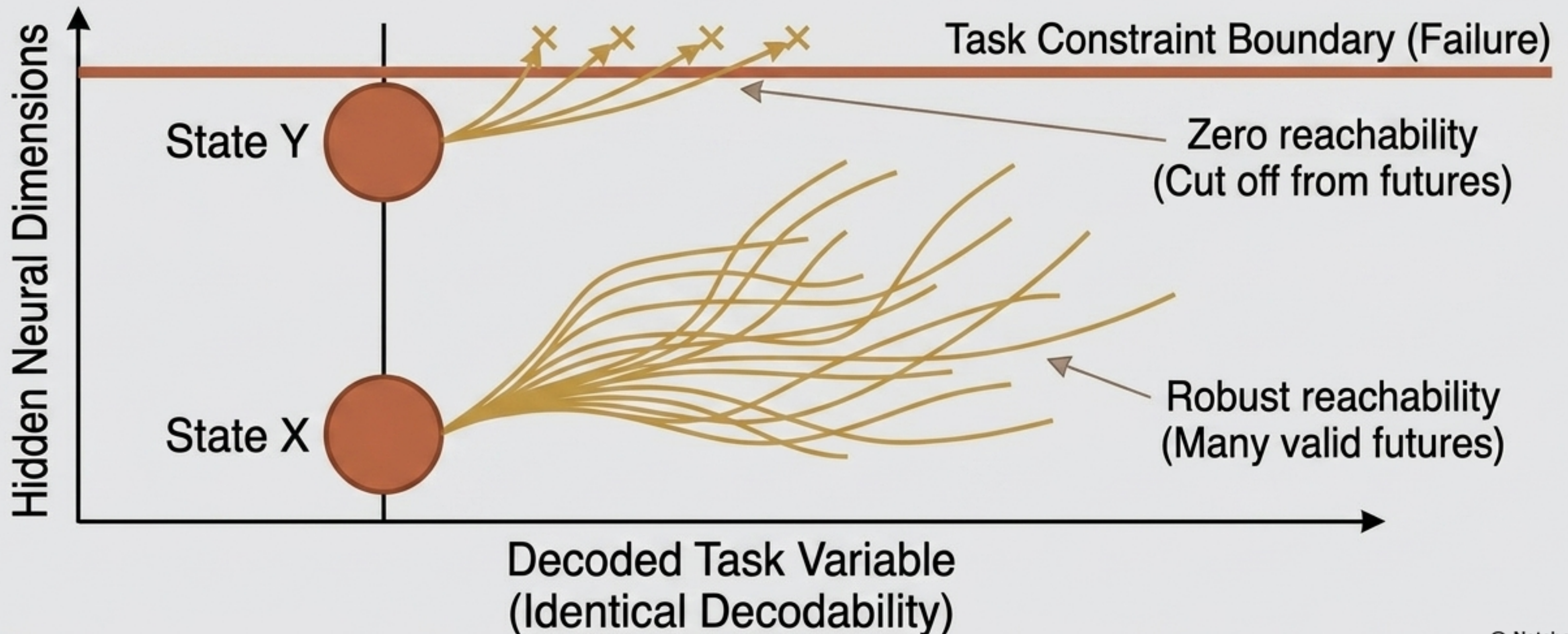
Current theories argue that a decodable readout remains stable in a null space even as raw geometry drifts. This explains the present (what can be read now) but fails to explain the future (what recovery trajectories remain possible).





# The Fundamental Difference Between Decoding and Reaching

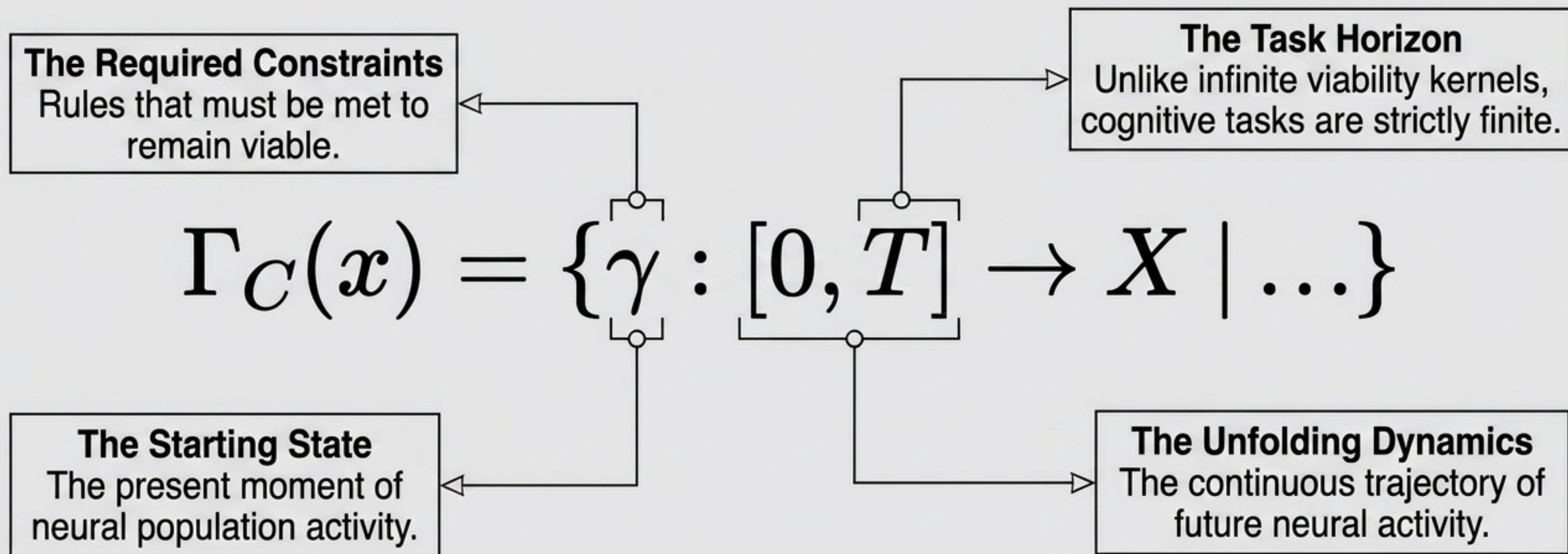
Two neural states can be identically decodable but have radically different behavioral futures. Decodability checks the present state; Reachability checks the envelope of viable trajectories.





# Identifying the True Invariant: Admissible Continuations

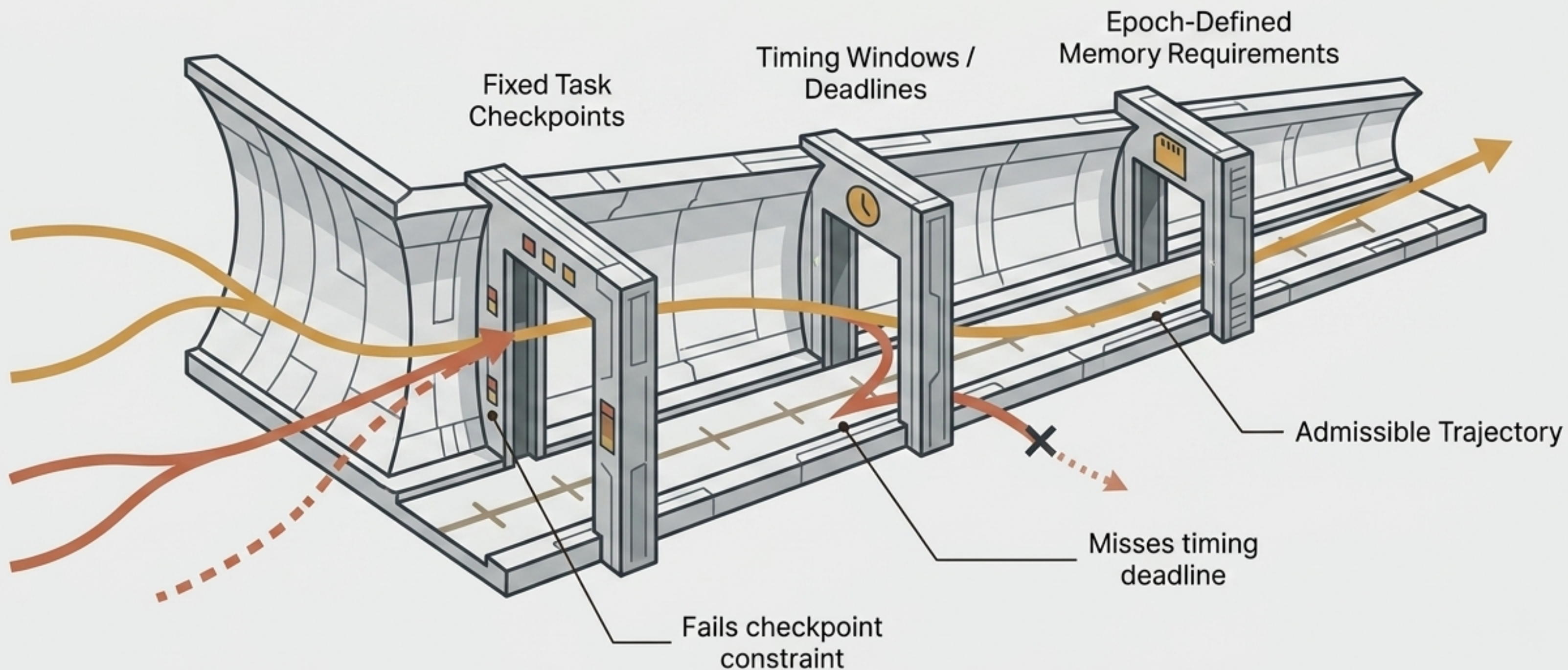
The minimal formal object needed to test reachability is the set of all future trajectories starting at a state that obey transition laws and satisfy ongoing task constraints.





# Constraining the Repertoire to Make It Falsifiable

Constraints (C) cannot just mean “trajectories that succeed.” They must be verifiable independent of final outcome, evaluated epoch by epoch before trial resolution.





# Parsing the Architectures of Neural Dynamics

Comparing what is conserved during learning, perturbation, and representational drift across the three competing architectures.

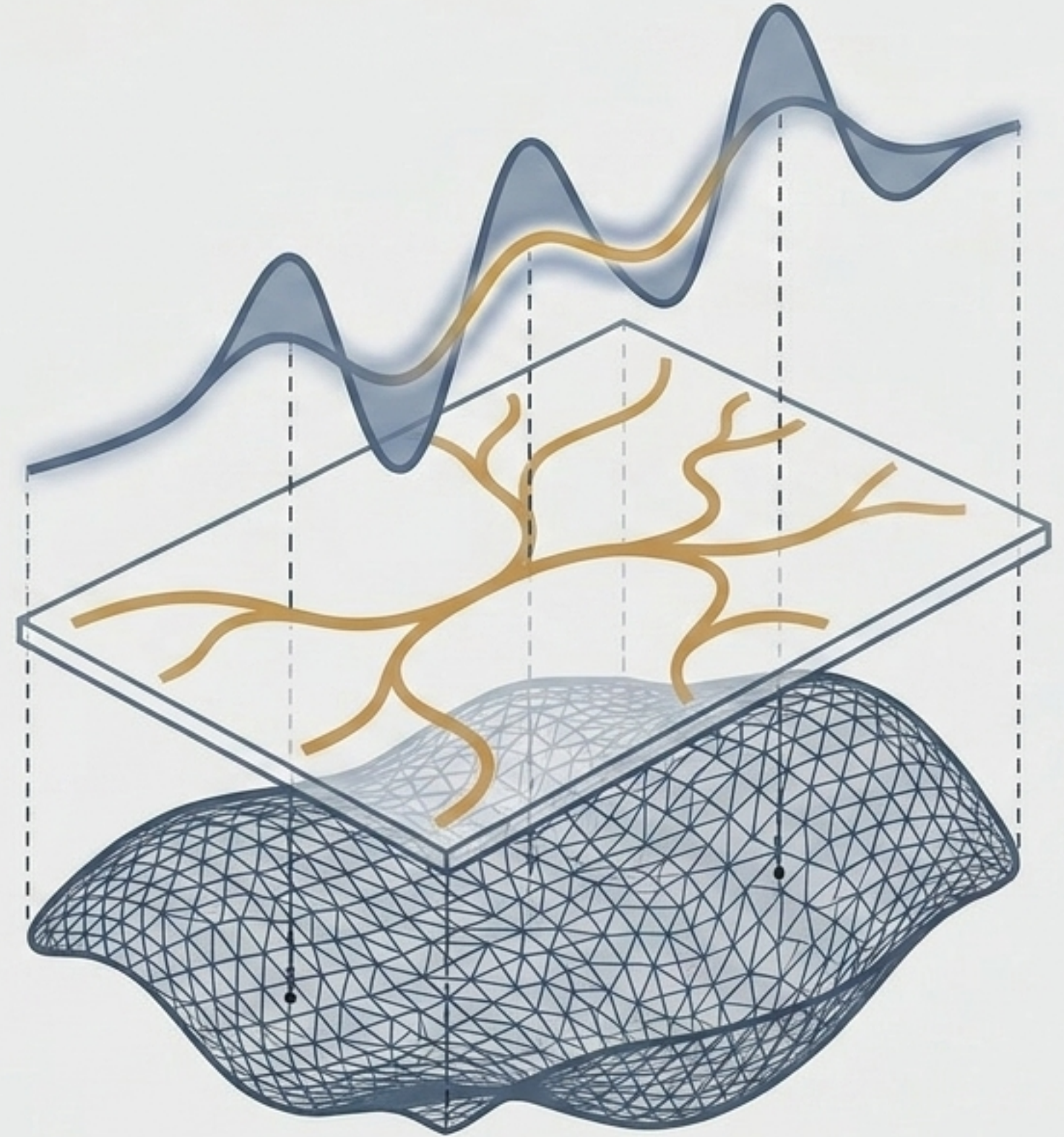
|                  | Manifold Geometry         | Decodable Readout      | Admissible Continuations   |
|------------------|---------------------------|------------------------|----------------------------|
| Core Question    | Covariance Structure      | Subspace Shadow        | Reachable Futures          |
| Evaluated Over   | Instantaneous Present     | Instantaneous Present  | Future Horizon             |
| Preserved During | Short-Term Perturbation   | Representational Drift | Drift & Recovery           |
| Failure Mode     | Forced outside covariance | Loss of readout signal | Cut off from valid futures |



# A Layered Architecture of Neural Control

These paradigms do not compete; they stack.  
Manifolds constrain the physical repertoire.  
Continuations define what remains reachable.  
Waves act as momentary selectors among  
those reachable paths.

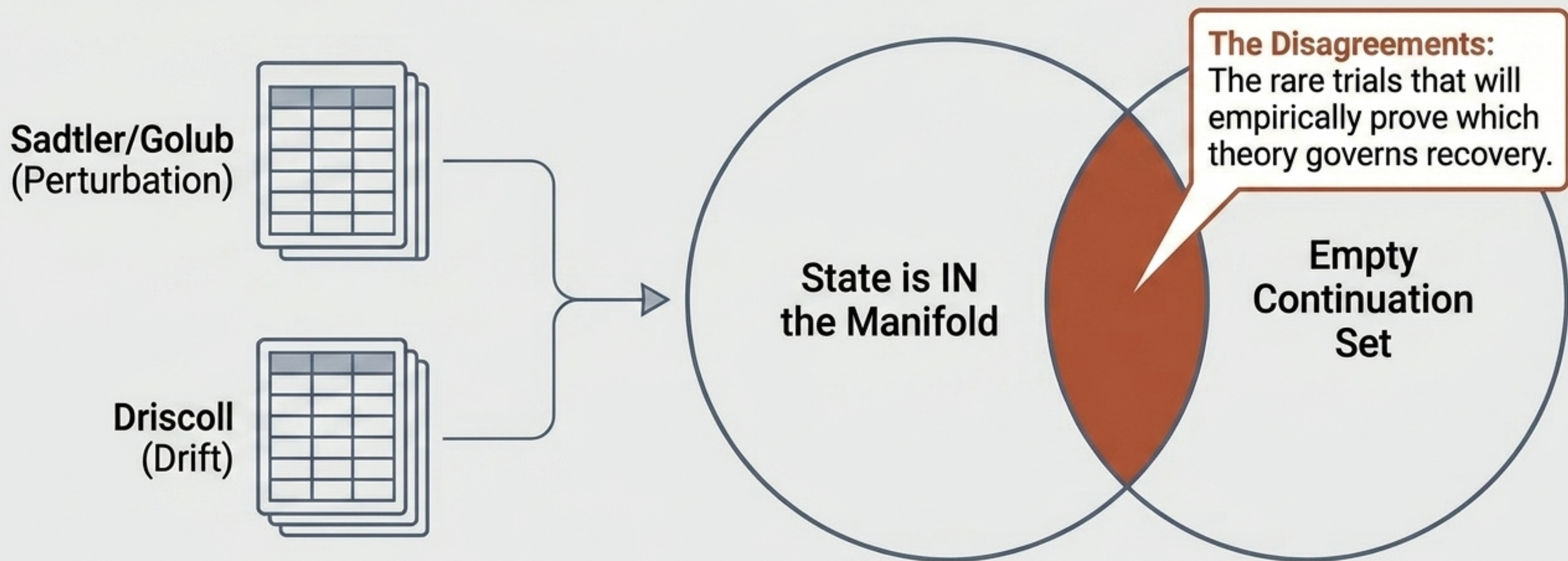
- Layer 3: Wave Dynamics**  
(Momentary Selector)
- Layer 2: Admissible Continuations**  
(Reachable Futures)
- Layer 1: The Manifold**  
(Anatomical Constraint)





# The Reanalysis Test: Hunting for Disagreements

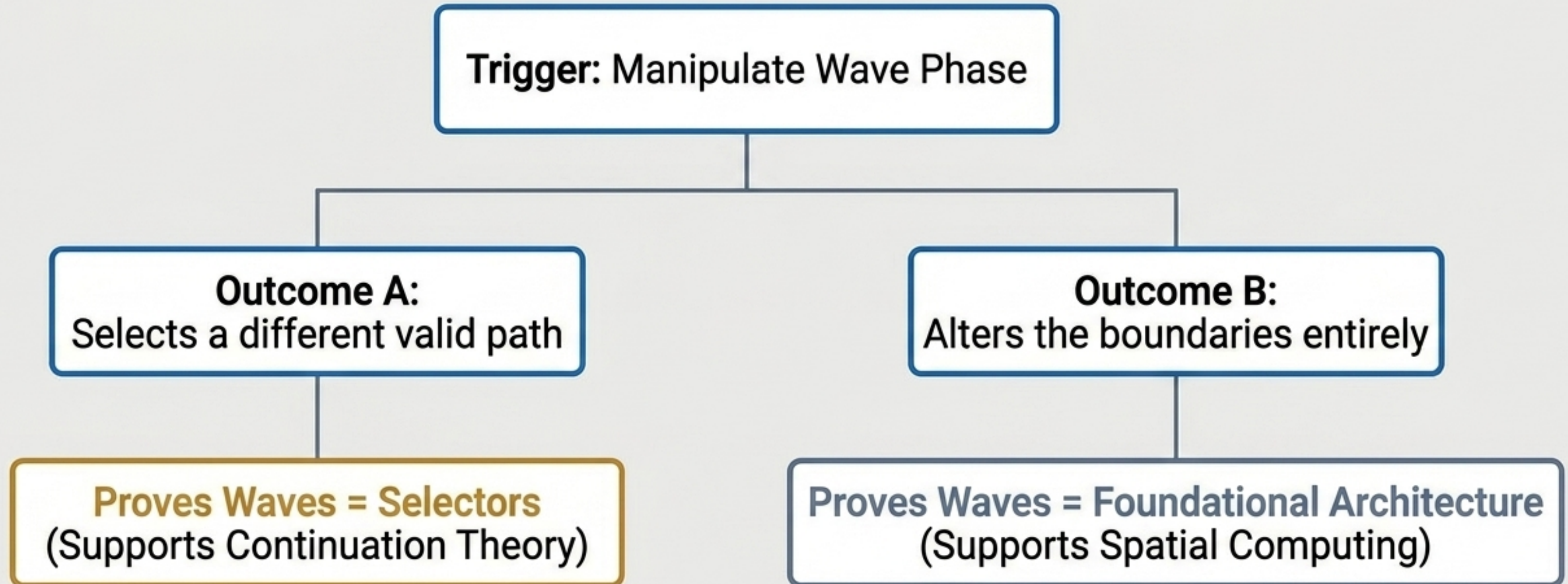
We can test this by reanalyzing existing perturbation (Sadtler/Golub) and drift (Driscoll) datasets. Do behavioral recoveries track continuation-set overlap better than manifold membership?





# The Decisive Experiment: Isolating Phase from Geometry

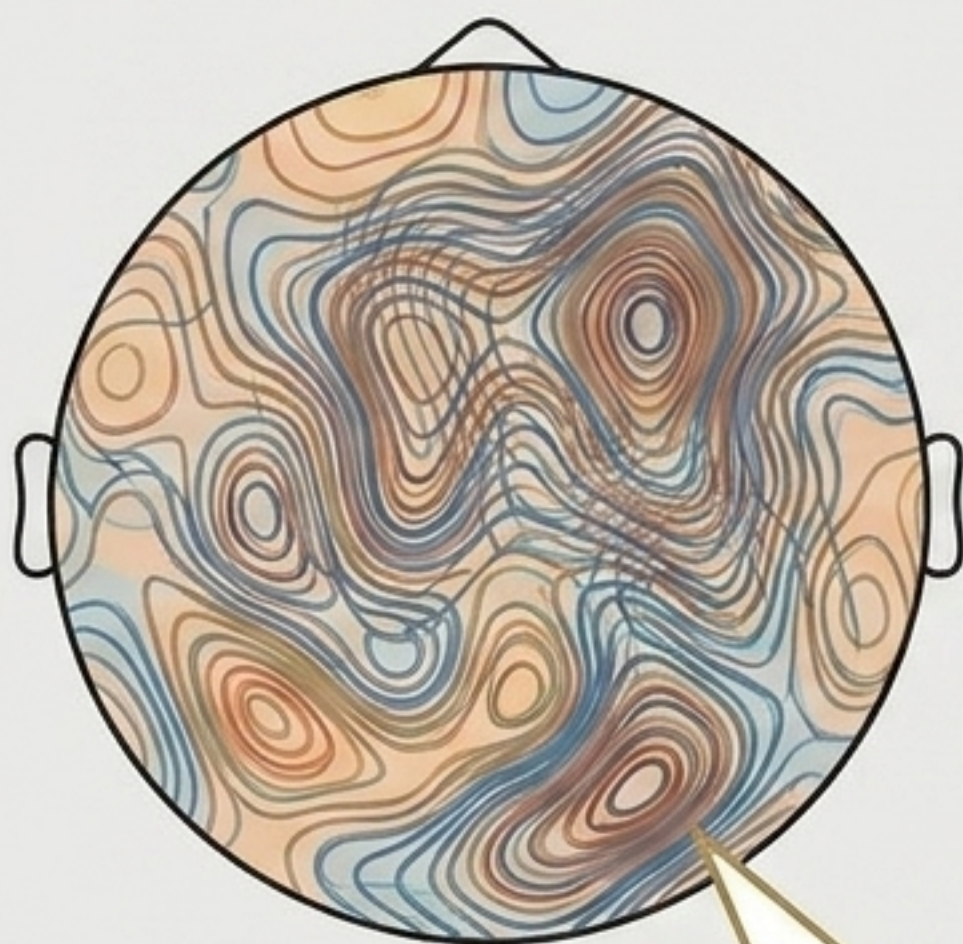
To prove wave necessity, we must independently manipulate oscillatory phase while holding population trajectory geometry and continuation sets constant.





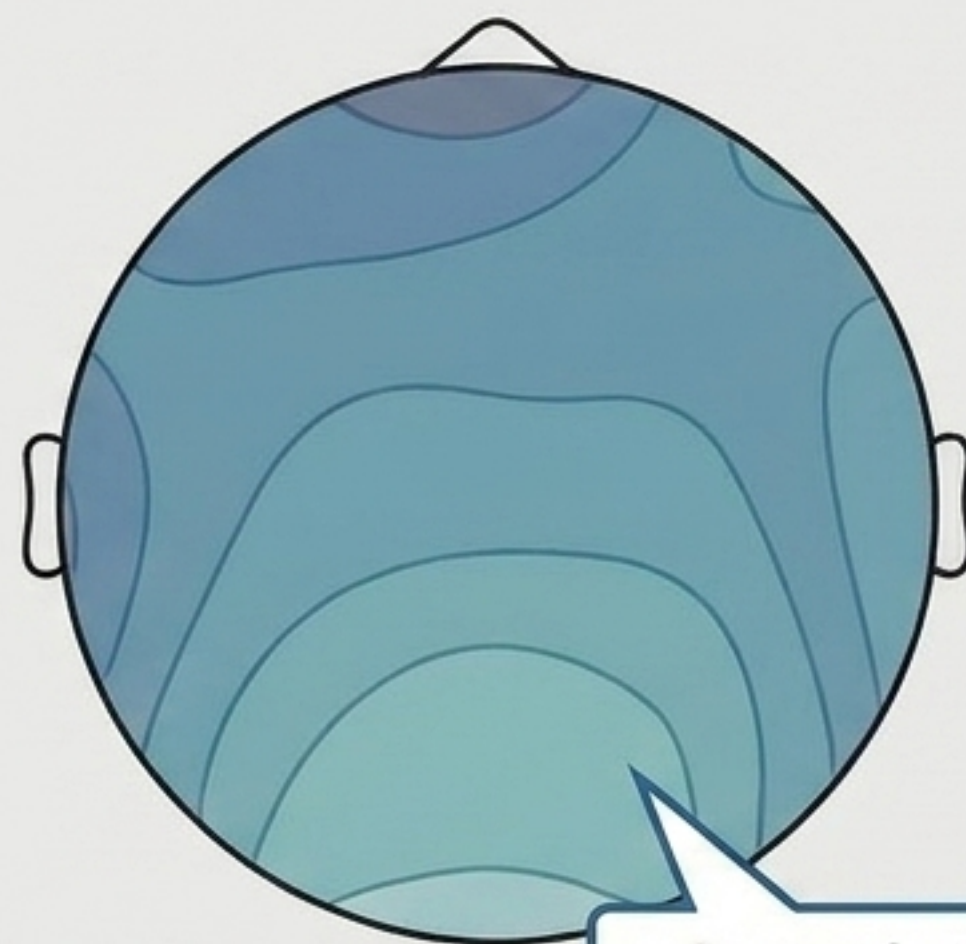
# Re-evaluating the Anesthesia Argument

Loss of consciousness corresponds to degraded wave Q-factors (Cabral et al.). But this proves waves define the structural boundaries of reachability, not that they act as rapid computational selectors.



Waking  
Eigenmodes

**Defining the Boundary**  
(Structural Reality)



Anesthetized  
Eigenmodes

**Loss of Repertoire**  
(Not a loss of a  
computational selector)



# Distinguishing What a State Encodes from What It Enables

The debate is not “Waves vs. No Waves.”

It is distinguishing **momentary readouts** from  
the envelope of reachable **futures**.

Identifying this **invariant** architecture, rather than its biological  
correlates, is the next frontier of consciousness theory.